THE ANALYST TOOLKIT

TOOLS MATTER, BUT SKILLS MATTER MORE

When it comes to analytics tools, **don't be afraid to specialize**; focus on building deep skills and expertise in a few key areas, rather than trying to master it all

*You have permission to **NOT** learn all of these tools*

THE ANALYST TOOLKIT

- **BI Platforms**
- Database Tools
- Data Prep/ETL Tools
- Spreadsheet Tools
- Programming Languages

BI PLATFORMS

Power BI | tableau | Qlik | SISENSE

Self-service tools designed to support the entire business intelligence workflow, including data prep, modeling, analysis, visualization and administration

INDUSTRY LEADERS*

- *Power BI*
- *Tableau*
- *Qlik*

TYPICAL USER ROLES

- *Business Intelligence Analyst*
- *Data Analyst*
- *Data Visualization Specialist*

COMMON USE CASES

- *Performance dashboards*
- *ETL & data prep*
- *Ad hoc data visualization*
- *Report administration*

PRO TIP: Rather than trying to everything at once, focus on going deep with one platform first; the skills you build will be highly transferrable if you need to switch

** Other popular BI platforms include Sisense, Domo, Google (Looker), and MicroStrategy*

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DATABASE TOOLS

MySQL | Microsoft SQL Server | PostgreSQL | mongoDB

Used for creating, managing, and querying information stored in relational database management systems (RDMS)

INDUSTRY LEADERS*

- *MySQL*
- *Microsoft SQL Server*
- *PostgreSQL*

TYPICAL USER ROLES

- *Database Administrator*
- *Business Intelligence Analyst*
- *Data Scientist*

COMMON USE CASES

- *Storing large datasets*
- *Ad hoc database analysis*
- *Producing custom tables or views for analysis and visualization*

PRO TIP: There are many "flavors" of SQL database tools, but they are all built on the same query language and universal standards (with slight variations in syntax)

** If you are looking for cloud database tools, consider tools like Amazon RDS/Redshift, Azure, MongoDB, etc.*

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Spreadsheet Tools
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DATA PREP/ETL TOOLS

talend Informatica alteryx

Used for extracting, cleaning, transforming, and loading data from disparate sources into a centralized location for analysis

INDUSTRY LEADERS*

- *Talend/Informatica (cloud)*
- *Alteryx*
- *Tableau Prep*
- *Power BI (Power Query)*

TYPICAL USER ROLES

- *Data Engineer*
- *Database Administrator*
- *Business Intelligence Analyst*

COMMON USE CASES

- *Building data pipelines for analysis*
- *Developing an automated ETL process for ongoing reporting*
- *Blending data from multiple sources into a central database or BI platform*

PRO TIP: Tableau Prep and Power Query are both excellent ETL tools, and are directly integrated with tools you're likely already using *(Tableau, Power BI & Excel)*

* While these are popular options, you can also find specialized ETL tools designed specifically for real-time/streaming, unstructured data, etc.

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BI Platforms
Database Tools
Data Prep/ETL Tools
Spreadsheet Tools
Programming Languages

SPREADSHEET TOOLS

Google Sheets

Used for creating, managing, modeling and analyzing structured data stored in rows and columns

INDUSTRY LEADERS*

- *Microsoft Excel*
- *Google Sheets*

TYPICAL USER ROLES

- *Data Analyst*
- *Business Intelligence Analyst*
- *Financial Analyst*
- *Anyone working with data*

COMMON USE CASES

- *Ad hoc analysis & visualization*
- *Financial modeling*
- *Forecasting & optimization*
- *Performance reporting*

PRO TIP: Excel might just be the most versatile tool in the analytics stack, and can be a great solution for anything from ad hoc analysis to complex data modeling

* There are other spreadsheet tools on the market (LibreOffice/Calc, Zoho, etc.), but Excel and Google Sheets are in a league of their own

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- **Programming Languages**

PROGRAMMING LANGUAGES

Coding languages and packages commonly used for statistical analysis, machine learning and data science

INDUSTRY LEADERS*

- *Python*
- *R*
- *SAS*
- *SQL*

TYPICAL USER ROLES

- *Data Scientist*
- *Machine Learning Engineer*
- *Data Engineer*

COMMON USE CASES

- *Training machine learning models*
- *Analyzing massive datasets*
- *Processing unstructured data*

PRO TIP: Most analytics roles won't require heavy programming unless you work in Data Science or Machine Learning; focus on Excel + SQL first, and build from there

** Python is a general-purpose language (also used for web dev, mobile apps, games, etc.), while R and SAS are geared towards statistical analysis*

COMMON TOOL STACKS

As you begin to develop your skills, it's important to identify a **collection *(or "stack")* of tools** which will help you perform the responsibilities of your specific role

BI ANALYST	DATA VIZ SPECIALIST	DATA ENGINEER	DATA SCIENTIST

There's no "right" or "wrong" stack, as long as the tools you use allow you to perform your job effectively

KEY TAKEAWAYS

Think about skills first, tools second

- *Don't focus on which tool you want to learn; focus on which SKILL you want to build*

Don't be afraid to specialize

- *Aim to go deep rather than broad, and focus on building 1-2 expert-level skills (vs. 10 mediocre ones)*

During the learning process, focus on one tool at a time

- *Staying focused helps you learn efficiently, retain your skills, and transfer your knowledge to similar tools*

Work towards building a "tool stack" specific to your role

- *For analytics professionals, we recommend Excel, SQL, and a BI platform like Power BI/Tableau*

TIPS FOR SUCCESS

Obsess over **OUTCOMES**

Master **SKILLS**, not tools

Don't be afraid to **SPECIALIZE**

Don't ignore the **SOFT SKILLS**

Never stop **LEARNING**

Follow your **OWN PATH**